

28/04/2025

CODE-A



Aakash

Medical | IIT-JEE | Foundations

Corporate Office: AESL, 3rd Floor, Incuspaze Campus-2, Plot-13,
Sector-18, Udyog Vihar, Gurugram, Haryana-122015



AIM - 720

(Advanced **INTENSIVE** Mastery for 720)

CC-115

CC-115

CC-115

CC-115

CC-115

CC-115

CC-115

MM : 720

CST-12

Time : 180 Mins.

Complete Syllabus of NEET

Instructions:

- Duration of Test is 3 hrs.
- The Test consists of 180 questions. The maximum marks are 720.
- There are three parts in the question paper consisting of Physics, Chemistry, Biology having 45 questions each in Physics and Chemistry and 90 questions in Biology).
- Each question carries +4 marks. For every wrong response, -1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
- Use blue/black ballpoint pen only to darken the appropriate circle.
- Mark should be dark and completely fill the circle.
- Dark only one circle for each entry.
- Dark the circle in the space provided only.
- Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

6 Aakashians scored a perfect 720/720 from Classroom Programs



720
720 | **Mridul M Anand**
3 Year Classroom



720
720 | **Ayush Naugraiya**
4 Year Classroom



720
720 | **Arghyadeep Dutta**
2 Year Classroom



720
720 | **Aryan Yadav**
1 Year Classroom



720
720 | **Palansha Agarwal**
2 Year Classroom



720
720 | **Iram Quazi**
1 Year Classroom

PHYSICS

1. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): The absolute error has the same unit as the quantity itself.

Reason (R): Fractional error has no unit.

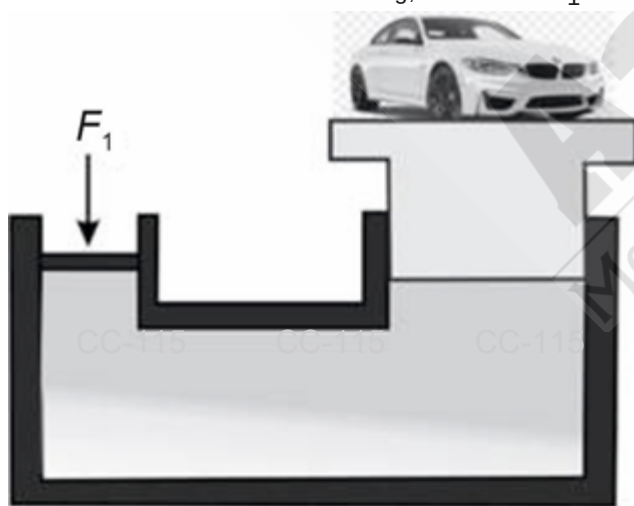
In the light of above statements, choose the correct answer from the options given below.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (3) (A) is true but (R) is false
- (4) Both (A) and (R) are false

2. A car accelerates from rest at constant rate of 2 m/s^2 for some time after which it decelerates at a constant rate of 3 m/s^2 to come to rest. If total time taken for the motion is 40 seconds then maximum velocity achieved by the car during motion is

- (1) 40 m/s
- (2) 48 m/s
- (3) 20 m/s
- (4) 60 m/s

3. In a car lift, compressed air exerts a force F_1 on a small piston having a radius of 10 cm. This pressure is transmitted to a large piston of radius 20 cm as shown in figure. If the mass of the car to be lifted is 1500 kg, the value of F_1 will be

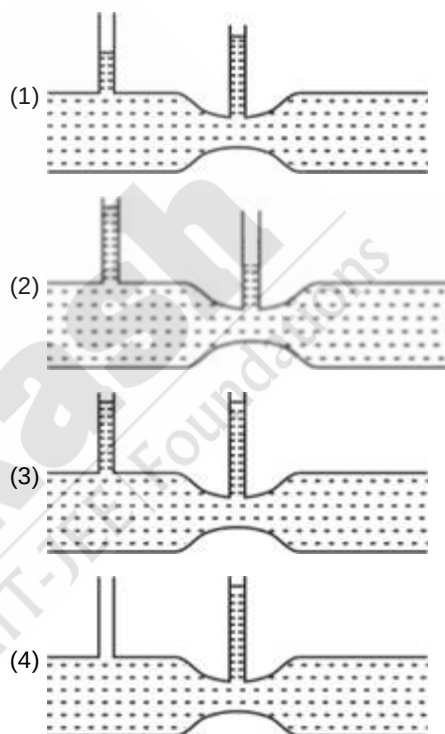


- (1) $1.5 \times 10^4 \text{ N}$
- (2) $1.5 \times 10^7 \text{ N}$
- (3) $3.75 \times 10^3 \text{ N}$
- (4) $7.5 \times 10^4 \text{ N}$

4. A vessel contains gas A at pressure P , volume V and temperature T . Another vessel contains gas B at pressure $2P$, volume $\frac{V}{2}$ and temperature $2T$. Ratio of number of molecules of B to A will be

- (1) $\frac{1}{2}$
- (2) 2
- (3) 4
- (4) $\frac{1}{4}$

5. Which of the following figures shown below is correct regarding the steady flow of an ideal liquid?



6. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): A digital circuit that follows logical relationship between the input and output voltages are generally known as logic gates.

Reason (R): NOR and NAND gates are not considered as universal gates.

In the light of above statements, choose the correct answer from the options given below.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (3) (A) is true but (R) is false
- (4) Both (A) and (R) are false

7. The mass of a proton is 1.0073 u and that of neutron is 1.0087 u (u = atomic mass unit). The binding energy of ${}^4_2\text{He}$ is (mass of helium nucleus = 4.0015 u)

(1) 28.4 MeV
 (2) 0.061 MeV
 (3) 3.05 MeV
 (4) 305 MeV

8. Consider the following statements:

A. Average power supplied to an inductor by an ac source over one complete cycle is zero.

B. Capacitive reactance is inversely proportional to both frequency of ac source and capacitance.

C. Power factor of AC series LCR circuit is equal to the product of resistance and impedance of the circuit.

The correct statement(s) is/are

(1) Both (B) and (C)
 (2) Both (A) and (B)
 (3) Both (A) and (C)
 (4) All (A), (B) and (C)

9. A bar magnet has length 3 cm, cross-sectional area 2 cm^2 and magnetic moment 3 A m^2 . The intensity of magnetisation of bar magnet is

(1) $2 \times 10^5\text{ A/m}$
 (2) $3 \times 10^5\text{ A/m}$
 (3) $4 \times 10^5\text{ A/m}$
 (4) $5 \times 10^5\text{ A/m}$

10. Consider statements (A) and (B) given below:

A. Thermodynamics deals with the process of conversion of heat into work only.

B. Heat given to a system and work done by the system are state variables in thermodynamics.

Choose the correct option.

(1) Statement (A) is correct but statement (B) is incorrect
 (2) Statement (A) is incorrect but statement (B) is correct
 (3) Both statements are correct
 (4) Both statements are incorrect

11. Match column I with column II.

Column-I (Types of EM waves)		Column-II (Production)
A. Infra-red	P	Rapid vibration of electrons in aerials
B. Radio	Q	Electrons in atoms emit light when they move from higher to lower energy level.
C. Light	R	Klystron valve
D. Microwave	S	Vibration of atoms and molecules

(1) A-P, B-R, C-S, D-Q

(2) A-S, B-P, C-Q, D-R

(3) A-Q, B-P, C-S, D-R

(4) A-S, B-R, C-P, D-Q

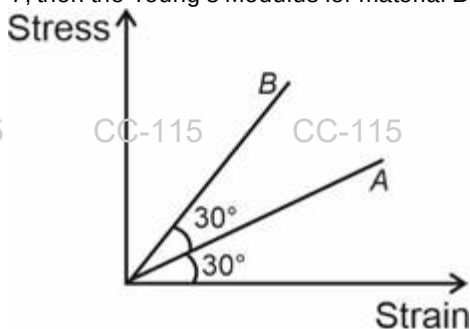
12. Two batteries one of emf 18 V and internal resistance $3\ \Omega$ while other of emf 12 V and internal resistance $2\ \Omega$ are connected in parallel with positive terminals together at one point and negative terminals together to other point. If across these points an ideal voltmeter is connected, the reading of voltmeter will be

(1) 16 V
 (2) 14.4 V
 (3) 15.3 V
 (4) 12.6 V

13. Two moles of an ideal monoatomic gas undergoes an adiabatic process from temperature 300 K to 600 K. Work done by this ideal gas in the process is

(1) 600R
 (2) -200R
 (3) -450R
 (4) -900R

14. In the graph shown, if the Young's Modulus of material A is Y, then the Young's Modulus for material B is



(1) $\sqrt{3}Y$
 (2) $Y/3$
 (3) $3Y$
 (4) $2Y$

15. A uniform circular disc of mass 2 kg and radius 100 cm (hinged at centre) is subjected to a constant torque of 4 Nm. If the disc was initially at rest, then its angular speed after 4 s will be

(1) 16 rad/s
(2) 4 rad/s
(3) 2 rad/s
(4) 8 rad/s

16. If potential energy is assumed to be zero at infinity, then

(1) The total energy of an orbiting satellite is negative of its potential energy.
(2) The potential energy of an orbiting satellite is twice of its total energy.
(3) The potential energy of an orbiting satellite is negative of its kinetic energy
(4) The total energy of an orbiting satellite is twice of its kinetic energy

17. A swimmer can swim with speed of 8 m/s in still water. 800 m wide river is flowing with speed of 4 m/s. Swimmer wants to cross the river in minimum time. Velocity of swimmer with respect to ground is (approximately)

(1) 9 m/s
(2) 10 m/s
(3) 12 m/s
(4) 5 m/s

18. Two masses each equal to M are moving on a circular path of radius R about a common centre. The gravitational force of attraction between the masses has magnitude

(1) $F = \frac{GM^2}{R^2}$
(2) $F = \frac{GM^2}{4R^2}$
(3) $F = \frac{4GM^2}{R^2}$
(4) $F = \frac{GM^2}{2R^2}$

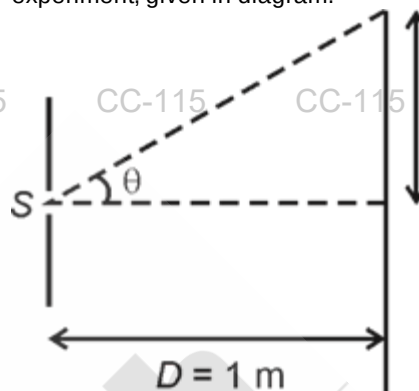
19. A string is fixed at both ends and the vibrations of string is given by the equation $y = 10\sin(2x)\cos(2t)$ where x, y are in cm and t is in second. Nearby node from left end at $x = 0$, is at a distance

(1) $\frac{\pi}{2}$ cm
(2) π cm
(3) 2 cm
(4) 4 cm

20. A steel wire 0.50 m long has a mass of 4.0×10^{-3} kg. If the wire is under a tension of 80 N, the speed of transverse waves on the wire is

(1) 93 m s^{-1}
(2) 100 m s^{-1}
(3) 50 m s^{-1}
(4) 98 m s^{-1}

21. Consider the experimental setup for single slit diffraction experiment, given in diagram.



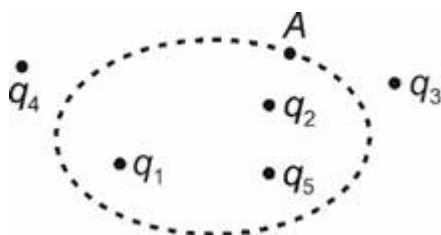
Wavelength of light is 800 nm. The first dark fringe obtained at angle θ is at a distance 0.8 mm from central bright. The width of central bright fringe is

(1) 0.8 mm
(2) 0.16 mm
(3) 1.6 mm
(4) 0.08 mm

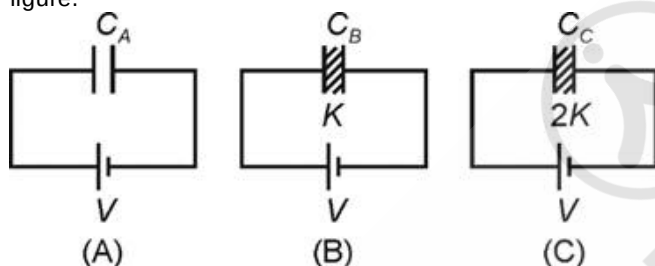
22. The sun delivers about 1400 watt/m^2 of electromagnetic flux to Earth's surface. If the flux falls on a roof of dimensions $8 \text{ m} \times 20 \text{ m}$ normally, then total power incident on the roof will be

(1) 224 kW
(2) 112 kW
(3) 32 kW
(4) 96 kW

23. A Gaussian surface encloses three of five positively charged particles as shown in the figure. The particles which contribute to the electric field at point A on the surface are



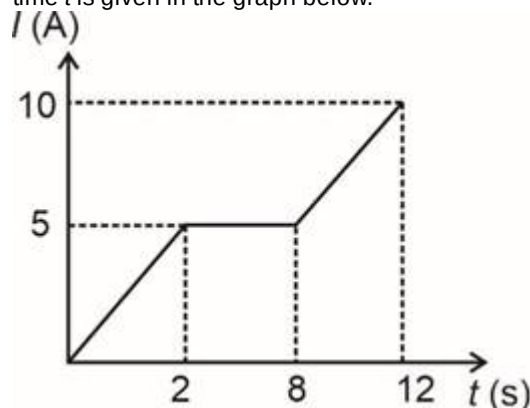
- (1) q_1 , q_2 and q_5 only
 (2) q_4 and q_5 only
 (3) q_3 , q_4 and q_5 only
 (4) q_1 , q_2 , q_3 , q_4 and q_5
24. Let us consider three initially uncharged identical capacitors having capacitance C . Space between the plates of two of the three capacitors is now filled with a di-electric material of di-electric constant K and $2K$ and each capacitor is now connected across identical batteries, as shown in figure.



The ratio of charge stored in all three capacitors in steady state will be

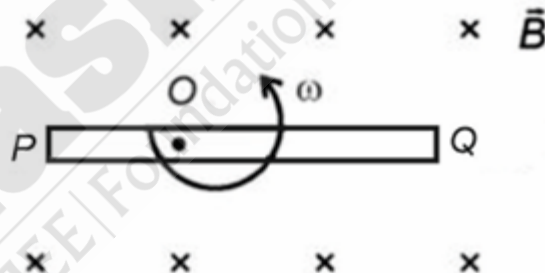
- (1) $1 : K : 2K$
 (2) $1 : 1 : 2K$
 (3) $1 : K : 3K$
 (4) $2K : 3K : 5K$

25. Variation of current I flowing through a conducting wire with time t is given in the graph below.



The amount of charge flowing through the wire in time interval $t = 2$ s to $t = 12$ s, will be

- (1) 20 C
 (2) 40 C
 (3) 60 C
 (4) 80 C
26. A conducting rod PQ of length $4L$ is rotating about a point O in a uniform magnetic field $\vec{B}$ directed into the paper as shown in the figure. Given that $PO = L$ and $OQ = 3L$, then



Potential difference between ends P and Q is

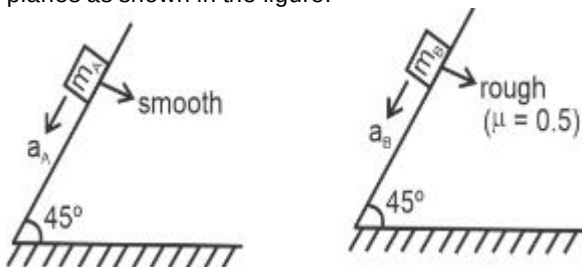
- (1) $\frac{1}{2} B\omega L^2$
 (2) $4B\omega L^2$
 (3) $\frac{1}{4} B\omega L^2$
 (4) $\frac{9}{2} B\omega L^2$
27. Consider a hypothetical hydrogen like atom. Wavelength of spectral lines corresponding to transition of electron from $n = m$ to $n = 1$ are given by

$$\lambda = \frac{1800m^2}{m^2 - 1} \text{ \AA} \quad (m = 2, 3, 4, 5, \dots, \infty)$$

Minimum energy of emitted photon is nearly

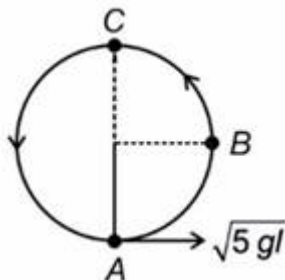
- (1) 5.2 eV
 (2) 7.03 eV
 (3) 10.2 eV
 (4) 13.6 eV

28. Two blocks A and B are released from rest on two inclined planes as shown in the figure.



The ratio of the accelerations (a_A/a_B) is

- (1) 1
(2) 2
(3) 1.5
(4) 0.8
29. A bob of mass m is suspended by light string of length l such that it is free to rotate in vertical circular plane as shown in the figure. At lowest position it is imparted a horizontal velocity $\sqrt{5gl}$ such that it just completes circular trajectory in vertical circle. Ratio of its KE at B and C will be



- (1) 2 : 1
(2) 3 : 1
(3) 5 : 3
(4) 3 : 2
30. Consider a system of two identical particles. One of the particles is at rest and the other has an acceleration a . Their centre of mass has an acceleration
- (1) Zero
(2) $\frac{a}{2}$
(3) a
(4) $2a$
31. The force acting on a particle is $(\hat{i} + 2\hat{j} + 3\hat{k})$ N. Find the torque (in N m) of this force about origin if position vector of the particle is $(7\hat{i} + 3\hat{j} + 5\hat{k})$ m.

- (1) $\hat{i} + 16\hat{j} - 11\hat{k}$
(2) $-\hat{i} - 16\hat{j} + 11\hat{k}$
(3) $\hat{i} + 16\hat{j} + 11\hat{k}$
(4) $-\hat{i} + 9\hat{j} + 11\hat{k}$

32. Which of the following example does not represent SHM?

- (1) Oscillations of a spring block system
(2) Motion of ball bearing inside smooth curved bowl, when released slightly away from equilibrium position
(3) Motion of oscillating mercury column in vertical U-tube
(4) Rotation of earth about its own axis

33. When 150 gm of ice at 0°C is mixed with 60 gm of water at 50°C in a container, the resulting temperature is

- (1) 2.7°C
(2) 3.7°C
(3) 5.7°C
(4) 0°C

34. Beta positive decay is

- (1) The conversion of neutron into proton along with neutrino and electron
(2) The conversion of proton into neutron along with antineutrino and electron
(3) The conversion of neutron into proton along with positron and antineutrino
(4) The conversion of proton into neutron along with neutrino and positron

35. In the I - V characteristics of a silicon p-n junction diode, the current varies from 10 mA to 20 mA when applied voltage varies from 1 V to 1.2 V in linear portion of forward biasing. The dynamic resistance of diode will be

- (1) 10 ohm
(2) 20 ohm
(3) 30 ohm
(4) 40 ohm

36. If E is the energy of n^{th} orbit of hydrogen atom, the energy of n^{th} orbit of He^+ ion will be

- (1) E
(2) $2E$
(3) $3E$
(4) $4E$

37. A glass sheet is grinded to form a double convex lens. The radius of curvature of two surfaces are 15 cm and 10 cm and the focal length of lens is 12 cm. Refractive index of glass sheet is

- (1) 1.2
(2) 1.4
(3) 1.5
(4) 1.6

38. In Young's double slit interference pattern, the fringe width increases with
- (1) Increasing slit to screen distance
 - (2) Increasing wavelength
 - (3) Decreasing slit width
 - (4) All of the above
39. Pick out wrong statement about the Kirchhoff's laws of electric circuit.
- (1) Outgoing currents adds up and are equal to incoming currents at a junction
 - (2) Starting with any point if we come back to same point, total potential change must be zero
 - (3) Junction rule is based on conservation of energy law
 - (4) Bending or reorienting the wire does not change the validity of Kirchhoff's junction rule.
40. A particle is projected with a velocity $\vec{v} = (3\hat{i} + 4\hat{j})$ m/s in the presence of uniform acceleration $\vec{a} = (4\hat{i} - 3\hat{j})$ m/s². The path of the particle will be
- (1) Straight line
 - (2) Parabolic
 - (3) Circular
 - (4) Elliptical
41. In projectile motion if air resistance (or any of such force opposing motion) is taken into consideration, then
- (1) Projectile would deviate from its idealised parabolic trajectory.
 - (2) Range would be less than that in absence of air.
 - (3) Maximum height attained would be greater than that in absence of air.
 - (4) Both (1) and (2) are correct.
42. n identical small drops of water having radius r coalesce to form a bigger drop. If surface tension of water is T then excess pressure in bigger drop will be
- (1) $\frac{n^3 4T}{r}$
 - (2) $\frac{2T}{\frac{1}{n^3 r}}$
 - (3) $\frac{4T}{nr}$
 - (4) $\frac{2T}{n^3 r}$
43. In an experiment on photoelectric emission for incident light of wavelength 1.98×10^{-7} m, stopping potential is found to be 2.5 V. What is maximum kinetic energy of emitted photoelectron?
- (1) 6.25 eV
 - (2) 2.5 eV
 - (3) 3.75 eV
 - (4) Zero
44. A particle of mass 4 kg is performing S.H.M. in a straight line under the action of force $F = (15 - 3x)$ N. Equilibrium position of particle will be
- (1) $x = 3$ m
 - (2) $x = 5$ m
 - (3) $x = 7$ m
 - (4) $x = 10$ m
45. A charge $q = 1.6 \times 10^{-12}$ C moving with speed of v m s⁻¹ crosses electric field $\vec{E} = 6 \times 10^4$ V m⁻¹ and magnetic field $\vec{B} = 1.2$ T. The electric field and magnetic fields are crossed and velocity v is also perpendicular to both. If the charge particle crosses both fields undeflected, the value of v is
- (1) 7.2×10^5
 - (2) 7.2×10^4
 - (3) 5×10^5
 - (4) 5×10^4

46. A hydrocarbon contains 80% carbon by mass. The empirical formula of the hydrocarbon is

- (1) CH
- (2) CH₂
- (3) CH₃
- (4) CH₄

47. Correct expression for Heisenberg uncertainty principle is

- (1) $\Delta x \times \Delta v_x \geq \frac{h}{4\pi}$
- (2) $\Delta x \times \Delta v_x \geq \frac{h}{4\pi m}$
- (3) $\Delta x \times \Delta v_x \geq \frac{h}{2\pi}$
- (4) $\Delta x \times \Delta v_x \geq \frac{h}{2\pi m}$

48. Which among the following set of quantum numbers is incorrect?

	n	l	m	s
(1)	4	2	-2	+1/2
(2)	3	1	+1	-1/2
(3)	5	3	-2	-1/2
(4)	2	1	-2	+1/2

- (1) (1)
- (2) (2)
- (3) (3)
- (4) (4)

49. Correct order of first ionization enthalpies is

- (1) Li < Be < B < C
- (2) Li < B < Be < C
- (3) Li < C < Be < B
- (4) Li < B < C < Be

50. Find out the true statement about electronegativity

- (1) It is always constant for given element
- (2) Electronegativity is inversely related to the metallic properties of elements
- (3) Electronegativity on mulliken scale = $\frac{IE_1 + EA_1}{2}$ where IE_1 and EA_1 are in eV atom⁻¹
- (4) Allred – Rochow scale is the most widely used to express electronegativity

51. Select the incorrect statement.

- (1) Resonance stabilizes the molecule
- (2) Resonance averages the bond characteristics as a whole
- (3) Resonance hybrid is least energetic structure
- (4) There is equilibrium between the canonical forms

52. Incorrect statement among the following is

- (1) Be₂ molecule does not exist
- (2) Boiling point of HF is higher than HCl
- (3) BrF₅ is trigonal bipyramidal in shape
- (4) Sulphur in SOCl₂ is sp³ hybridised

53. If $\Delta_f H^\circ$ of CaCO₃(s), CaO(s) and CO₂(g) are -1200, -630 and -400 kJ mol⁻¹ respectively then $\Delta_r H^\circ$ of the reaction CaCO₃(s) → CaO(s) + CO₂(g) is

- (1) -170 kJ mol⁻¹
- (2) 170 kJ mol⁻¹
- (3) -2230 kJ mol⁻¹
- (4) 2230 kJ mol⁻¹

54. For a reaction CaCO₃(s) ⇌ CaO(s) + CO₂(g) ($K_C = 0.5$ molL⁻¹), the maximum moles of CO₂(g) formed at equilibrium in 5 L container is

- (1) 5
- (2) 2.5
- (3) 1.25
- (4) 10

55. Which is not correct match of disproportionation product of given elements?

	Reactants	Products
(1)	Hydrogen peroxide	Water and oxygen
(2)	Octa sulphur in alkaline medium	Sulphide and thiosulphate ions
(3)	Phosphorous in alkaline medium	Phosgene and oxyanion of phosphorous of +1 oxidation state
(4)	Chlorine in alkaline medium	Chloride and hypochlorite ions

- (1) (1)
- (2) (2)
- (3) (3)
- (4) (4)

56. Products obtained when SiO₂ reacts with NaOH and HF respectively are

- (1) Na₂SiO₃ and SiF₂
- (2) Na₂Si₂O₇ and SiF₄
- (3) Na₂SiO₃ and SiF₄
- (4) Na₂Si₂O₇ and Na₂SiF₆

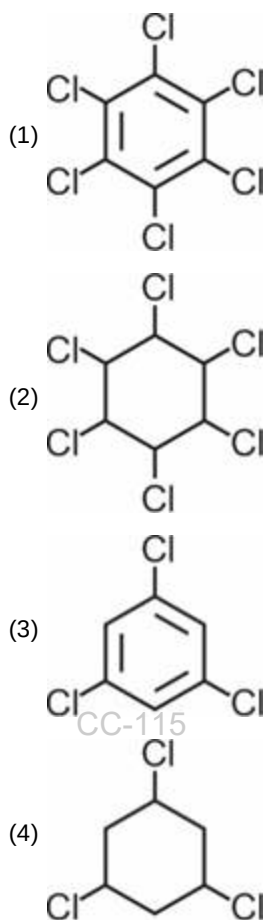
57. Incorrect statement among the following is

- (1) Tert-butyl carbocation is more stable than benzyl carbocation
- (2) Neopentyl carbocation does not show hyperconjugation
- (3) Total number of resonating structures of $\text{CH}_2=\text{CH}-\text{CHO}$ is 2
- (4) $\text{CH}_2=\text{CH}-\dot{\text{C}}\text{H}_2$ is more stable than $\text{C}_6\text{H}_5\dot{\text{C}}\text{H}_2$

58. During estimation of nitrogen present in an organic compound by Kjeldahl's method the ammonia evolved from 0.4 g of compound is neutralised by 10 mL of 1 M sulphuric acid solution. The percentage of nitrogen in the compound is

- (1) 56 %
- (2) 28 %
- (3) 70 %
- (4) 35 %

59. Structure of gammaxane is



60. Molality of an aqueous solution of urea in which mole fraction of urea is 0.2, is

- (1) 12.33 m
- (2) 13.88 m
- (3) 10.33 m
- (4) 11.23 m

61. In Daniell cell of Zn and Cu electrodes, if applied external opposite voltage is more than 1.1 V then select the correct option

- (a) No flow of current is observed
- (b) Flow of electrons is from Zn electrode to Cu electrode
- (c) Cell is acting as electrolytic cell
- (d) Zinc electrode is acting as cathode while copper acting as anode

- (1) Only a
- (2) Only b
- (3) Only c and d
- (4) Only b, c and d

62. Mechanism of hypothetical reaction $\text{X}_2 + \text{Y}_2 \rightarrow 2\text{XY}$, is given below

- (i) $\text{X}_2 \rightleftharpoons 2\text{X}$ (fast)
- (ii) $\text{X} + \text{Y}_2 \rightarrow \text{XY} + \text{Y}$ (slow)
- (iii) $\text{X} + \text{Y} \rightarrow \text{XY}$ (fast)

Select the correct statements among a, b, c and d

- (a) Order w.r.t. X_2 is 1
- (b) Order w.r.t. Y_2 is 0.5
- (c) Overall order of reaction is 1.5
- (d) Order w.r.t. X is 0.5

- (1) Only c
- (2) Only a and c
- (3) Only b and d
- (4) Only c and d

63. The correct order against the indicated property is

- (1) Melting point : $\text{SbH}_3 > \text{AsH}_3 > \text{PH}_3 > \text{NH}_3$

- (2) Boiling point : $\text{SbH}_3 > \text{NH}_3 > \text{AsH}_3 > \text{PH}_3$

- (3) Bond angle : $\text{PH}_3 > \text{NH}_3 > \text{SbH}_3 > \text{AsH}_3$

- (4) Bond energy : $\text{NH}_3 > \text{PH}_3 > \text{SbH}_3 > \text{AsH}_3$

64. The shape of XeOF_4 , XeO_3 and XeF_4 respectively are

- (1) Square pyramidal, Trigonal planar, see saw
- (2) Tetrahedral, Pyramidal, see-saw
- (3) Square pyramidal, Trigonal pyramidal, square planar
- (4) Octahedral, Trigonal pyramidal, tetrahedral

65. In alkaline medium MnO_4^- oxidises I^- into

- (1) IO_4^-
- (2) IO_2^-
- (3) I_2
- (4) IO_3^-

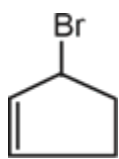
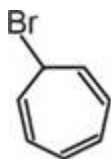
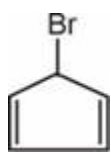
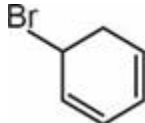
66. Which of the following pairs of elements exhibit similar radii?

- (1) Zr, Hf
- (2) Ti, Zr
- (3) Cu, Ag
- (4) Mn, Tc

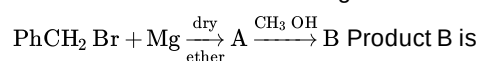
67. The incorrect statement regarding crystal field theory is

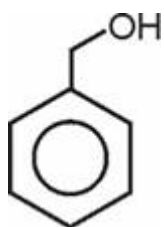
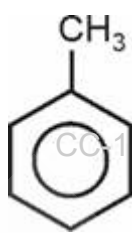
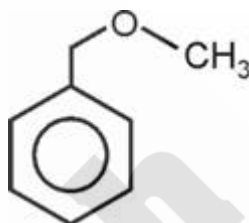
- (1) It considered the metal-ligand bond to be purely ionic
- (2) The pattern of splitting depends upon the nature of the crystal field
- (3) The degeneracy of five d -orbitals is lifted when negative field of ligands surrounds the metal atom/ion
- (4) In octahedral complexes, the orbitals $d_{x^2-y^2}$ and d_{z^2} will be lowered in energy relative to average energy in the spherical crystal field

68. Among the following which bromide will react fastest with AgNO_3 solution

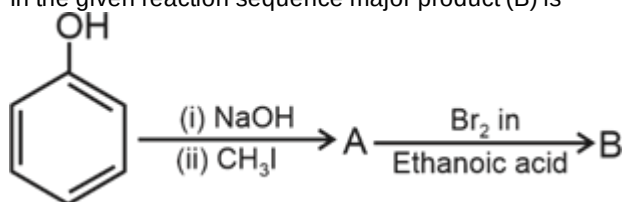
- (1) 
- (2) 
- (3) 
- (4) 

69. Consider the following reaction sequence



- (1) 
- (2) 
- (3) 
- (4) 

70. In the given reaction sequence major product (B) is



- (1)
- (2)
- (3)
- (4)

71. The ether which is most difficult to cleave on heating with HBr is

- (1)
- (2) Ph-CH₂-O-CH₂-Ph
- (3) Ph-O-CH₃
- (4) Ph-O-Ph

72. Given below are two statements one is Assertion (A) other is Reason (R).

Assertion (A) : $2p_z$ orbital of one atom cannot combine with the $2p_x$ or $2p_y$ orbital of the another atom.

Reason (R) : $2p_z$ orbital and $2p_x$ or $2p_y$ orbitals have different orientation.

In the light of above statements, choose the correct answer from the options given below.

- (1) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (2) (A) is true but (R) is false
- (3) (A) is false but (R) is true
- (4) Both (A) and (R) are true and (R) is the correct explanation of (A)

73. Given below are two statements one is Assertion (A) other is Reason (R).

Assertion (A) : For an ideal gas undergoing adiabatic process, $\Delta U = W$.

Reason (R) : In an adiabatic process, no heat is transferred between the system and surrounding.

In the light of above statements, choose the correct answer from the options given below.

- (1) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (2) (A) is true but (R) is false
- (3) (A) is false but (R) is true
- (4) Both (A) and (R) are true and (R) is the correct explanation of (A)

74. Solution of 0.1 N NH₄OH and 0.01 N NH₄Cl has pH 10.25. Then the pK_b of NH₄OH at 25°C is

- (1) 5.75
- (2) 3.75
- (3) 4.75
- (4) 6.75

75. Given below are two statements one is Assertion (A) other is Reason (R).

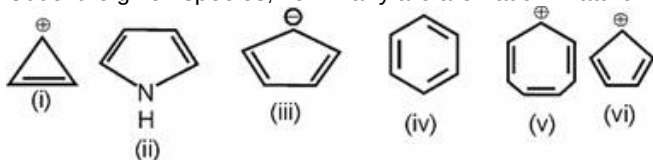
Assertion (A) : The t-butyl free radical is more stable than ethyl free radical.

Reason (R) : The t-butyl free radical is stabilised by greater +H and more +I.

In the light of above statements, choose the correct answer from the options given below.

- (1) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (2) (A) is true but (R) is false
- (3) (A) is false but (R) is true
- (4) Both (A) and (R) are true and (R) is the correct explanation of (A)

76. Out of the given species, how many are aromatic in nature?



- (1) 5
(2) 6
(3) 4
(4) 3

77. Select the incorrect statement among following

When concentration of electrolyte for equation

- (1) $\Lambda_m = \Lambda_m^0 - A\sqrt{C}$ approaches zero, the molar conductivity is known as limiting molar conductivity
(2) The value of constant 'A' is independent of solvent and temperature for a given electrolyte
(3) The value of constant 'A' for NaCl and KCl are same
(4) The variation of Λ_m for strong and weak electrolytes in a given solvent and temperature are different

78. Given below are two statements.

Statement (I): $t_{1/2}$ for a zero order reaction is directly proportional to the initial concentration of the reactants.

Statement (II): For a zero order reaction, $t_{1/2} = \frac{A_0}{k}$. (A_0 = Initial concentration)

In the light of above statements, choose the most appropriate answer from the options given below.

- (1) Both statement I and Statement II are correct
(2) Both statement I and Statement II are incorrect
(3) Statement I is incorrect but statement II is correct
(4) Statement I is correct but statement II is incorrect

79. Which among the following oxides is anhydride of nitric acid?

- (1) N_2O_3
(2) NO_2
(3) N_2O_5
(4) N_2O

80. For which of the following complexes, the C-O bond length is maximum?

- (1) $[Mn(CO)_6]^+$
(2) $[Ni(CO)_4]$
(3) $[Cr(CO)_6]$
(4) $[V(CO)_6]^+$

81. Given below are two statements.

Statement (I) : o-Nitrophenol is more acidic than p-Nitrophenol.

Statement (II) : pK_a of o-Nitrophenol is lower than m-Nitrophenol.

In the light of above statements, choose the most appropriate answer from the options given below.

- (1) Both statement I and Statement II are correct
(2) Both statement I and Statement II are incorrect
(3) Statement I is incorrect but statement II is correct
(4) Statement I is correct but statement II is incorrect

82. Given below are two statements.

Statement (I): Benzaldehyde is more reactive than propanal in nucleophilic addition reaction.

Statement (II): Carbonyl carbon of benzaldehyde is more electrophilic than carbonyl carbon of propanone.

In the light of above statements, choose the most appropriate answer from the options given below.

- (1) Both statement I and Statement II are correct
(2) Both statement I and Statement II are incorrect
(3) Statement I is incorrect but statement II is correct
(4) Statement I is correct but statement II is incorrect

83. Given below are two statements.

Statement (I): Aniline is ortho/para directing in nature.

Statement (II): Anilinium ion is meta directing.

In the light of above statements, choose the most appropriate answer from the options given below.

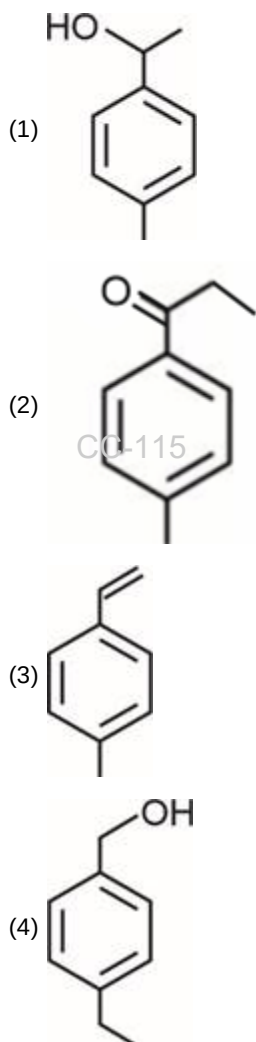
- (1) Both statement I and Statement II are correct
(2) Both statement I and Statement II are incorrect
(3) Statement I is incorrect but statement II is correct
(4) Statement I is correct but statement II is incorrect

84. Match the name of vitamins given in column I with their deficiency diseases given in column II

Column I	Column II
a. Thiamine	(i) Scurvy
b. Riboflavin	(ii) Beri Beri
c. Pyridoxine	(iii) Cheilosis
d. Ascorbic acid	(iv) Convulsions

- (1) a(ii), b(iii), c(iv), d(i)
(2) a(iii), b(ii), c(iv), d(i)
(3) a(iv), b(iii), c(ii), d(i)
(4) a(iv), b(iii), c(i), d(ii)

85. Which among the following will give positive haloform test?



86. 2 moles and 3 moles of liquids, A ($P_A^\circ = 120$ torr) and B ($P_B^\circ = 80$ torr) respectively are mixed to form an ideal solution. The mole fraction of B in vapour phase will be

- (1) $\frac{2}{5}$
 (2) $\frac{1}{2}$
 (3) $\frac{1}{4}$
 (4) $\frac{3}{5}$

87. Complete hydrolysis of RNA gives

- (1) β -D-ribose and phosphoric acid only
 (2) β -D-2-deoxyribose and phosphoric acid only
 (3) β -D-ribose, phosphoric acid and nitrogen containing heterocyclic compounds
 (4) β -D-2-deoxyribose and nitrogen containing heterocyclic compounds only

88. Match amines in List-I with their pK_b in List-II

List-I	List-II
a. Methanamine	(i) 3.00
b. N-Methylmethanamine	(ii) 3.38
c. N, N-Dimethylmethanamine	(iii) 3.27
d. N-Ethylethanamine	(iv) 4.22

Correct match is

- (1) a(ii), b(iii), c(iv), d(i)
 (2) a(iv), b(ii), c(iii), d(i)
 (3) a(i), b(iii), c(ii), d(iv)
 (4) a(iv), b(iii), c(i), d(ii)

89. Match cations in List-I with their group reagent in List-II

List-I	List-II
a. Cu^{2+}	(i) $(\text{NH}_4)_2\text{CO}_3 + \text{NH}_4\text{OH}$
b. Ba^{2+}	(ii) $\text{NH}_4\text{OH} + \text{NH}_4\text{Cl}$
c. Al^{3+}	(iii) $\text{H}_2\text{S} + \text{NH}_4\text{OH}$
d. Ni^{2+}	(iv) $\text{HCl} + \text{H}_2\text{S}$

Correct match is

- (1) a(ii), b(iii), c(i), d(iv)
 (2) a(iii), b(i), c(ii), d(iv)
 (3) a(iv), b(i), c(ii), d(iii)
 (4) a(iv), b(ii), c(i), d(iii)

90. Which among the following salt solution will give rotten egg like smell on addition of dilute H_2SO_4 .

- (1) NaCl(aq)
 (2) $\text{Na}_2\text{S(aq)}$
 (3) $\text{KNO}_3\text{(aq)}$
 (4) $\text{Na}_2\text{CO}_3\text{(aq)}$

BIOLOGY

91. Between which of the following taxa, there might be greater difficulty of determining relationship?
- (1) *Mangifera* and *Triticum*
 - (2) Dicotyledonae and Monocotyledonae
 - (3) Sapindales and Poales
 - (4) Convolvulaceae and Solanaceae
92. Read the following Assertion (A) and Reason (R) statements and select the **correct** option.
Assertion (A): Deuteromycetes are commonly known as imperfect fungi.
Reason (R): Sexual reproduction is absent in the members of Deuteromycetes.
- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
 - (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
 - (3) Both (A) and (R) are false
 - (4) (A) is true but (R) is false
93. A protozoan without locomotory structure is
- (1) *Entamoeba*
 - (2) *Trypanosoma*
 - (3) *Plasmodium*
 - (4) *Paramecium*
94. Read the following statements and select the **correct** option.
Statement A: Bacteria as a group shows the most extensive metabolic diversity.
Statement B: Bacteria may be photosynthetic autotrophic or chemosynthetic autotrophic.
- (1) Only statement A is correct
 - (2) Only statement B is correct
 - (3) Both statements A and B are correct
 - (4) Both statements A and B are incorrect
95. The records or estimates regarding biodiversity do not give any figures for prokaryotes because
- a. Conventional taxonomic methods are not suitable for identifying microbial species.
 - b. Many prokaryotic species are simply not culturable under laboratory conditions.
 - c. They are not seen with naked eyes.
 - d. They do not show sexual reproduction which is essential to increase the biodiversity.
- The **correct** ones are
- (1) a and b
 - (2) b and c
 - (3) c and d
 - (4) a and d
96. Select the **incorrect** match.
- (1) *Ectocarpus* – Filamentous
 - (2) Kelps – Possess carotenoids
 - (3) *Chlamydomonas* – Cup shaped chloroplast
 - (4) *Chara* – Colonial green alga
- (1) (1)
 - (2) (2)
 - (3) (3)
 - (4) (4)
97. Which of the following plants shows zygomorphic flower with imbricate aestivation of petals?
- (1) *Calotropis*
 - (2) Cotton
 - (3) *Datura*
 - (4) Gulmohur
98. The ornamental plant of Solanaceae family is
- (1) Lupin
 - (2) *Petunia*
 - (3) Tulip
 - (4) *Gloriosa*
99. Read the following statements and choose the **correct** option.
Statement A: In all pteridophytes, the female gametophytes are retained on the parent sporophyte for variable periods.
Statement B: In the members of pteridophytes, male gametes are released from the antheridia, to the mouth of archegonium.
- (1) Only statement A is correct
 - (2) Only statement B is correct
 - (3) Both statements A and B are correct
 - (4) Both statements A and B are incorrect
100. Collenchyma is also a living simple tissue just like parenchyma. However, it differs from the latter as it
- (1) Contains cells that may be oval, spherical or polygonal.
 - (2) Has small intercellular spaces.
 - (3) Has pectin rich thickened corners of the cell wall and provides mechanical support to growing parts of plants.
 - (4) Can perform photosynthesis.

101. Select the **correct** statements with respect to bulliform cells.

- a. Modified abaxial hypodermal cells.
- b. Large, empty, colourless cells.
- c. Photosynthetic, contain large number of chloroplasts.
- d. Make the leaves curl inwards during water stress conditions.
- e. Help to minimise water loss.

- (1) Only a, b and d
- (2) Only a, c and e
- (3) Only b, d and e
- (4) All a, b, c, d and e

102. Axillary buds

- a. Are found in axil of leaves
- b. Are capable of forming a flower or tendril
- c. Are derived from shoot apical meristem
- d. Are secondary meristem

The **correct** ones are

- (1) a and b only
- (2) b and c only
- (3) a, b and c only
- (4) All a, b, c and d

103. The membrane bound cell organelles which are involved in protein synthesis and its modifications are respectively

- (1) ER and Golgi body
- (2) ER and ribosome
- (3) Golgi body and lysosome
- (4) Mitochondrion and ribosome

104. The ribosomes of chloroplast are

- a. 70S type
 - b. Larger than cytoplasmic ribosome
 - c. Made up of 50S & 40S subunits
- The **correct** one(s) is/are

- (1) a only
- (2) a and c only
- (3) b and c only
- (4) All a, b and c

105. Match the following columns w.r.t. chromosomes and select the **correct** option.

	Column I		Column II
a.	Centromere	(i)	Disc shaped structure
b.	Satellite	(ii)	Primary constriction
c.	Kinetochore	(iii)	Basic protein
d.	Histone	(iv)	Small fragment that appears due to secondary constriction

- (1) a(iii), b(iv), c(i), d(ii)
- (2) a(ii), b(iv), c(i), d(iii)
- (3) a(ii), b(i), c(iv), d(iii)
- (4) a(iii), b(i), c(iv), d(ii)

106. Fibreless intercellular matrix is the characteristic feature of

- (1) Areolar tissue
- (2) Blood
- (3) Cartilage
- (4) Adipose tissue

107. Which of the following is a primary metabolite?

- (1) Carotenoid
- (2) Anthocyanin
- (3) Cellulose
- (4) Ribose

108. The analysis of outer hard covering of the cockroach reveals the presence of a polysaccharide with N-acetyl glucosamine as its monomeric units. Identify the polysaccharide and select the **correct** option w.r.t it

- (1) It is a structural polysaccharide
- (2) It is also found in exoskeleton of annelids
- (3) It is a derivative of galactose
- (4) It is the most abundant polysaccharide in nature

109. A portion of which among the following acts as a common passage for air and food in case of humans?

- (1) Larynx
- (2) Pharynx
- (3) Trachea
- (4) Bronchi

110. A special system of blood vessels in our body that is exclusively meant for the circulation of blood to and from the cardiac musculature is

- (1) Pulmonary circulation
- (2) Coronary circulation
- (3) Renal portal system
- (4) Hepatic portal system

111. The organic compound 'X' that forms 10-15% of the total cellular mass in living organisms, is made up of monomeric units that are joined together with the help of

- (1) Peptide bonds
- (2) Glycosidic bonds
- (3) Phosphodiester bonds
- (4) Hydrogen bonds

112. Which among the following is a heteropolymer?

- (1) Cellulose
- (2) Starch
- (3) Collagen
- (4) Glycogen

113. In humans, the value of pCO_2 in alveolar air is

- (1) Equal to the value of pCO_2 in systemic veins
- (2) More than the value of pO_2 in systemic arteries
- (3) Less than the value of pO_2 in pulmonary vein
- (4) Equal to the value of pCO_2 in pulmonary artery

114. Select the incorrect statement from the following w.r.t. cockroach.

- (1) A pair of spermatheca is present in the 6th abdominal segment in the female cockroach.
- (2) In male cockroach, the ejaculatory duct opens into male gonopore situated ventral to anus.

The female reproductive system consists of two large ovaries, lying laterally in the 6th to 8th abdominal segments.

Male reproductive system consists of a pair of testes one lying on each lateral side in the 4th -6th abdominal segments.

115. A 55 years old female met with a mild accident that resulted in multiple fractures as her bone mass has decreased due to ageing. The most likely reason for decreased bone mass could be

- (1) Normal estrogen levels in the blood
- (2) Low PTH in blood
- (3) Low calcium in bones due to low estrogen
- (4) Increased progesterone in the blood

116. Which of the given cell organelles is common to both prokaryotic as well as eukaryotic cells?

- (1) Golgi apparatus
- (2) Ribosome
- (3) Plastid
- (4) Mitochondrion

117. When a host fish species becomes extinct, its unique assemblage of parasites also meets the same fate. This can be explained as an example of

- (1) Alien species invasion
- (2) Mutualism
- (3) Co-extinction
- (4) Endemism

118. Read the following statements and select the **correct** option.

Statement A: Mitosis is seen in both diploid and haploid cells.

Statement B: Meiosis involves two sequential cycles of nuclear and cell division but only a single cycle of DNA replication.

- (1) Only statement A is correct
- (2) Only statement B is correct
- (3) Both statements A and B are incorrect
- (4) Both statements A and B are correct

119. Read the following statements and identify them as true (T) or false (F) and accordingly choose the **correct** option.

A. Splitting of centromere takes place during anaphase of mitosis.

B. During metaphase of mitosis, each chromatid of a chromosome remains connected to pole of its side by spindle fibres.

C. Nucleolus, Golgi complex and ER reform during the end of prophase.

D. In animal cells, cytokinesis is carried out by furrow formation in plasma membrane.

	A	B	C	D
(1)	T	T	F	F
(2)	T	T	F	T
(3)	F	T	T	T
(4)	T	F	F	T

- (1) (1)
- (2) (2)
- (3) (3)
- (4) (4)

120. G₁ phase corresponds to the interval between mitosis and initiation of

- (1) DNA replication
- (2) G₂ phase
- (3) G₀ phase
- (4) Post mitotic phase

121. Which of the following is **true** about carotenoids?

- (1) Serve as major primary pigment in the photosynthetic machinery
- (2) Form the reaction center of the photosystem
- (3) Prevent reaction center from photo-oxidation
- (4) Show blue color in chromatogram

122. Mark the **incorrect** statement w.r.t. C₄ plants.

- (1) They have a special type of leaf anatomy.
- (2) They can tolerate higher temperatures.
- (3) They lack a wasteful process called photo- respiration.
- (4) They show lower productivity and minimum yield as compared to the C₃ plants.

123. Mark the **incorrect** statement about aerobic respiration.

- (1) Most common in higher organisms
- (2) It involves complete oxidation of organic substances
- (3) It does not occur in any prokaryote
- (4) Glycolysis is the first step of aerobic respiration and it does not involve release of CO₂.

124. Substrate level phosphorylation in glycolysis takes place during the conversion of

- (1) 2-phosphoglycerate to 3-PGA
- (2) 2-phosphoglycerate to PEP
- (3) PEP to pyruvic acid
- (4) Glucose-6-phosphate to fructose-6-phosphate

125. Developmental and environmental heterophylly are exhibited by A and B , respectively.

Select the **correct** option to fill in the blanks A and B.

- (1) A – Coriander, B – Cotton
- (2) A – Cotton, B – Buttercup
- (3) A – Buttercup, B – Coriander
- (4) A – Cotton, B – Larkspur

126. Identify these statements as true (T) or false (F) and select the **correct** option.

- A - Gibberellins induce early seed production in pines.
 B - Several naturally occurring substances like NAA and kinetin promote cell division.
 C - Lateral shoot growth is promoted by cytokinin.

- (1) A(F), B(T), C(T)
- (2) A(T), B(F), C(F)
- (3) A(F), B(T), C(F)
- (4) A(T), B(F), C(T)

127. Select the **incorrect** statement regarding the viability period of pollen grains.

- (1) It depends upon the chromosome number of a species.
- (2) In some cereals the viability is lost within 30 minutes of their release.
- (3) Viability of pollen grains can be increased by storing them at very low temperature.
- (4) In some leguminous plants, pollen grains maintain viability for months.

128. The radicle and the root cap in monocot seeds are enclosed in an undifferentiated sheath called the

- (1) Coleoptile
- (2) Coleorhiza
- (3) Epicotyl
- (4) Scutellum

129. Select the characteristics **true** for wind pollinated flowers and mark the option accordingly.

- i. Light and non-sticky pollen grains.
- ii. Well exposed stamens.
- iii. Large feathery stigma.
- iv. Numerous ovules in each ovary.
- v. Mucilagenous covering around pollen grains.

- (1) iii, iv and v only
- (2) i, ii and iv only
- (3) ii, iii and iv only
- (4) i, ii and iii only

130. If a hybrid tall pea plant is selfed, what percent of F₁ individuals would be of true breeding line regarding the plant height?

- (1) 50%
- (2) 12.5%
- (3) 100%
- (4) 25%

131. Choose the **correct** option to complete the analogy w.r.t. muscle contraction.

Mitochondria : ATP :: Sarcoplasmic reticulum : ———

- (1) Ca^{2+}
- (2) Mg^{2+}
- (3) Fats
- (4) Proteins

132. Select the correct match/representation w.r.t a healthy adult man.

(1)	Percentage of monocytes in total WBCs	=	Percentage of proteins in plasma
(2)	Percentage of neutrophils in total WBCs	=	Percentage of water in plasma blood
(3)	Percentage of plasma in blood	=	Percentage of eosinophils in total WBCs
(4)	Percentage of formed elements in blood	=	Percentage of lymphocytes in total WBCs

- (1) (1)
- (2) (2)
- (3) (3)
- (4) (4)

133. Choose the incorrect statement.

- (1) Neurons are excitable cells because their membranes are in a polarized state.
The channels present on the neural membrane are selectively permeable for different ions and negatively charged larger proteins.
- (2) During the polarized state of a neuron, there are more Na^+ outside the cell than inside.
- (3) Ionic gradient across the resting membrane is maintained by the sodium – potassium pump.

134. What happens to GFR due to constrictory effect of ADH on blood vessels?

- (1) Increases
- (2) Decreases
- (3) Tends to remain at the same level
- (4) First increases then decreases

135. The hormones which stimulate erythropoiesis are secreted by all of the given endocrine glands, **except**

- (1) Adrenal cortex
- (2) Testis
- (3) Thyroid gland
- (4) Parathyroid gland

136. Which of the following is considered as the anterior most part of *Balanoglossus*?

- (1) Proboscis
- (2) Collar
- (3) Trunk
- (4) Stomochord

137. The excited repolarised nerve fibre becomes once more responsive for further stimulation within fraction of a second due to

- (1) Rapid efflux of K^+
- (2) Rise in permeability to Na^+ across its membrane
- (3) Decrease in permeability of Ca^{2+} across its membrane
- (4) Rise in the permeability to Mg^{2+}

138. Read the following statements and choose the **correct** option.

- (a) According to Thomas Malthus, populations are stable in size except for seasonal fluctuations.
- (b) When we describe the story of life on Earth, we treat evolution as a consequence of a process called natural selection.
- (c) Fitness is the end result of the ability to adapt and get selected by nature.

- (1) (a) and (b) are incorrect, and (c) is correct
- (2) (a) and (c) are correct, and (b) is incorrect
- (3) (b) and (c) are correct, and (a) is incorrect
- (4) (a), (b) and (c) are correct

139. A and B are the two key concepts of Darwinian theory of evolution.

Identify A and B and select the **correct** option.

	A	B
(1)	Branching descent	Mutation
(2)	Natural selection	Saltation
(3)	Natural selection	Branching descent
(4)	Founder effect	Saltation

- (1) (1)
- (2) (2)
- (3) (3)
- (4) (4)

140.How many of the mentioned hormones are not secreted by the organised endocrine glands of our body?

PRL, T₄, ANF, Epinephrine, CCK, Testosterone, Erythropoietin, Thymosin

Choose the **correct** option.

- (1) Three
- (2) Four
- (3) Five
- (4) Two

141.Which of the following is not true according to Hugo deVries?

- (1) Evolution is a gradual process
- (2) Mutation causes speciation
- (3) Mutations are random and directionless
- (4) Saltation can cause speciation

142.In humans, cortical nephrons differ from juxta-medullary nephrons in all, **except**

- (1) Number
- (2) Length of loop of Henle
- (3) Length of vasa recta
- (4) Presence of peritubular capillaries

143.In agarose gel electrophoresis, DNA fragments are separated mainly on the basis of

- a. Size
 - b. Charge
 - c. Charge to size ratio
 - d. Length of agarose gel
- Select the **correct** option.

- (1) a only
- (2) b only
- (3) a, b and c
- (4) a, b, c and d

144.Select the **correct** sequence w.r.t. evolution of plant forms.

- (1) *Rhynia*-type plants → *Psilophyton* → Seed ferns → Progymnosperms → Cycads
- (2) Chlorophyte ancestors → *Rhynia*-type plants → Tracheophyte ancestors → *Psilophyton*
- (3) Tracheophyte ancestors → *Zosterophyllum* → Lycopods
- (4) Chlorophyte ancestors → Tracheophyte ancestors → Bryophytes

145.Choose the odd one w.r.t. the structures that provides the site of interaction of lymphocytes with the antigen, which then proliferate to become effector cells.

- (1) Peyer's patches
- (2) Bone marrow
- (3) MALT
- (4) Spleen

146.Select the **correct** match in the table given below w.r.t. comparison between humans and frogs.

	Structure/system	Human	Frog
(1)	Gland	Thyroid as well as parathyroid	Only thyroid
(2)	Cerebral hemisphere	Paired	Unpaired
(3)	Hepatic portal system	Present	Absent
(4)	Pons	Present	Absent

(1) (1)

(2) (2)

(3) (3)

(4) (4)

147.One of the core techniques of biotechnology which involves maintenance of sterile ambience in chemical engineering is

- (1) Genetic engineering
- (2) Bioprocess engineering
- (3) Gene therapy
- (4) Biofortification

148.Identify the structures labelled as X, Y and Z in the given figure.



Subunit of which of the following binds to the calcium to expose myosin-binding site on actin filament?

- (1) X only
- (2) Y and Z
- (3) X and Z
- (4) Z only

149.The ART that involves the transfer of an ovum collected from a donor into the fallopian tube of another female who cannot produce one, but can provide suitable environment for fertilisation and further development, is

- (1) ZIFT
- (2) GIFT
- (3) IUI
- (4) IVF-ET

150.The exchange of material between blood and interstitial fluid occurs only at the level of

- (1) Arteries
- (2) Veins
- (3) Capillaries
- (4) Arterioles

151. Read the following statements and choose the **correct** option.

Statement A : Allergy is due to the release of chemicals like histamine and serotonin from the mast cells.

Statement B: The use of drugs like anti-histamine, adrenaline and steroids can quickly reduce the symptoms of allergy.

- (1) Both statements A and B are incorrect
- (2) Both statements A and B are correct
- (3) Only statement A is correct
- (4) Only statement B is correct

152. Match the columns and select the **correct** option.

	Column I		Column II
a.	Gastrin	(i)	Inhibits gastric secretion
b.	Gastric inhibitory peptide	(ii)	Stimulates the secretion of pancreatic enzymes
c.	Secretin	(iii)	Stimulates the secretion of water and bicarbonate from pancreas
d.	Cholecystokinin	(iv)	Stimulates the secretion of HCl

- (1) a(ii), b(iii), c(i), d(iv)
- (2) a(iv), b(i), c(iii), d(ii)
- (3) a(ii), b(i), c(iii), d(iv)
- (4) a(i), b(ii), c(iv), d(iii)

153. If you create a GM crop, which agency of Government of India will you approach to get permission for its commercial use?

- (1) IARI
- (2) CDRI
- (3) GEAC
- (4) WHO

154. Choose the **incorrect** statement w.r.t sponges.

- (1) Sponges reproduce asexually by fragmentation.
- (2) Larval stages are morphologically distinct from adults.
- (3) The process of digestion occurs in spongocoel.
- (4) Water enters into the body through ostia and leaves through osculum.

155. In a plasmid, a foreign DNA is ligated within the coding sequence of β -galactosidase. In the presence of a chromogenic substrate, recombinants will produce _____.

Select the **correct** option to fill in the blank.

- (1) Blue coloured colonies
- (2) White coloured colonies
- (3) Pink coloured colonies
- (4) No colonies

156. If we want to choose a species for experimental verification of chromosomal theory of inheritance, then on which of the following features would we focus that must be present in this species?

- (a) It can be grown in laboratory.
- (b) It should have large number of chromosomes.
- (c) It should produce large number of progenies in a single mating.
- (d) Male and female individuals can be easily distinguished.
- (e) It should also reproduce through asexual means.

The **correct** ones are

- (1) (b), (d) and (e)
- (2) (b), (c) and (d)
- (3) (a), (c) and (d)
- (4) (a), (b), (c) and (e)

157. Which of the following statements is **incorrect** w.r.t. sickle-cell anaemia?

Substitution of amino acid in the globin protein results

- (1) due to single base substitution at the sixth codon of the beta globin gene from GAG to GUG.
- (2) The defect is caused by the substitution of valine (val) by glutamic acid (Glu).
- (3) The mutant haemoglobin molecule undergoes polymerization under low oxygen tension.
- (4) Shape of RBC changes from biconcave disc to elongated sickle like structure.

158. In case of _____ the F_1 generation resembles both the parents. Choose the **correct** option to fill in the blank.

- (1) Incomplete dominance
- (2) Co-dominance
- (3) Complete dominance
- (4) Pleiotropy

159. The main enzyme for DNA replication is

- (1) DNA dependent DNA polymerase
- (2) DNA dependent RNA polymerase
- (3) RNA dependent DNA polymerase
- (4) RNA dependent RNA polymerase

160.In the experiments conducted by Griffith on *Streptococcus pneumoniae*, it was observed that

- (1) Mice injected with non-pathogenic strain become resistant to pathogenic stain
- (2) Heat-killed pathogenic strain is responsible for making non-pathogenic bacterium to pathogenic one
- (3) Non-pathogenic strain when mixed with pathogenic strain, the latter is also converted into non-pathogenic
- (4) Mutant strain of bacterium converts the mice into mutant type

161.Who proposed the Central Dogma in molecular biology which states that the genetic information flows from DNA → RNA → Protein?

- (1) James Watson
- (2) Maurice Wilkins
- (3) Francis Crick
- (4) Erwin Chargaff

162.Steps included in DNA fingerprinting are as follows:

- (A) Hybridisation using labelled VNTR probe, and detection of hybridised DNA fragments by autoradiography
- (B) Digestion of DNA by restriction endonucleases
- (C) Separation of DNA fragments by electrophoresis
- (D) Transferring (blotting) of separated DNA fragments to synthetic membranes, such as nitrocellulose or nylon
- (E) Isolation of DNA

Correct sequence of the steps in DNA fingerprinting technique is

- (1) (E) → (B) → (C) → (D) → (A)
- (2) (B) → (C) → (D) → (E) → (A)
- (3) (C) → (B) → (E) → (D) → (A)
- (4) (D) → (B) → (C) → (A) → (E)

163.Select the **incorrectly** matched pair.

- (1) mRNA – Provides template for protein synthesis
- (2) tRNA – Has the catalytic capability and read the genetic code directly from DNA
- (3) rRNA – Plays the structural and catalytic role during the translation
- (4) Sigma factor – An important factor for the initiation of RNA synthesis

164.A plant is affected with several root pathogens. Which of the following organisms should be employed for its treatment?

- (1) Ladybird
- (2) *Trichoderma*
- (3) Dragonflies
- (4) *Bacillus thuringiensis*

165.One of the objectives of Ganga Action plan and Yamuna Action Plan is to

- (1) Build large dams to produce hydroelectric energy, so that pollution from thermal power plants can be reduced
- (2) Build a large number of STPs so that only treated sewage may be discharged in the river
- (3) Divert the river flow so that during rainy seasons flood can be prevented
- (4) Introduce different species of aquatic plants and animals so that the health of aquatic ecosystem can be maintained

166.The interaction in which one species benefits and the other is neither harmed nor benefitted is seen in

- (1) Mycorrhizae associations between fungi and roots of higher plants.
- (2) Sea anemone and clown fish
- (3) Abingdon tortoise and goats in Galapagos island
- (4) *Cuscuta* on hedge plants

167.Population size of a given area or space can be determined by many ways. One of the methods called total number count is not useful for

- (1) Siberian cranes at Bharatpur wetlands
- (2) Humans living in city Delhi
- (3) Dense culture of bacteria in a petri dish
- (4) Trees in a particular small area

168.The integral form of the exponential growth equation is:

- (1) $\frac{dN}{dt} = N$
- (2) $\frac{dN}{dt} = rN$
- (3) $N_t = N_0 e^{rt}$
- (4) $N_0 = N_t r^{et}$

169.The biomass of an ecosystem is generally upright. It means the biomass will be greatest at the bottom. Exception to this can be for

- (1) Tree ecosystem
- (2) Grassland ecosystem
- (3) Sea ecosystem
- (4) Forest ecosystem

170. Consider the following four conditions (a-d) and select the **correct** pair of them as adaptation to environment in Kangaroo rat.

- (a) Meets all its water requirement through its internal fat oxidation.
- (b) Basks in sun when temperature is low.
- (c) Ability to concentrate its urine.
- (d) Presence of thick layer of fat below the skin.

- (1) (a) and (b)
- (2) (b) and (c)
- (3) (c) and (d)
- (4) (a) and (c)

171. How many approximate varieties of Basmati rice are documented in India?

- (1) 27
- (2) 30
- (3) 12
- (4) 2,00,000

172. Select the **correct** sequence for the steps involved in a cycle of PCR.

- (1) Denaturation → Annealing → Extension
- (2) Annealing → Denaturation → Extension
- (3) Denaturation → Extension → Annealing
- (4) Extension → Annealing → Denaturation

173. Enzyme which catalyses hydrolysis of peptide bond belongs to which class?

- (1) Oxidoreductases
- (2) Hydrolases
- (3) Lyases
- (4) Ligases

174. Gland involved in production of hormone which is responsible for childbirth and milk ejection is _____. Choose the **correct** option to fill in the blank.

- (1) Posterior pituitary
- (2) Anterior pituitary
- (3) Hypothalamus
- (4) Placenta

175. Which of the following represents the correct recognition sequence for *EcoR* I restriction endonuclease in one strand of DNA?

- (1) 5'-GGATTC3'
- (2) 5'-GAATTC-3'
- (3) 5'-GGCC3'
- (4) 5'-CCTGG-3'

176. Early detection of diseases is **not** possible by performing

- (1) RDT
- (2) Urine analysis
- (3) PCR
- (4) ELISA

177. Mode of action of oral steroidal contraceptive pills includes all, **except**

- (1) Inhibition of sperm motility but enhancement of their fertilization ability
- (2) Altering the quality of cervical mucus
- (3) Inhibition of ovulation
- (4) Inhibition of implantation

178. In human heart, the maximum number of action potentials are generated by

- (1) SAN
- (2) AV bundle
- (3) AVN
- (4) Purkinje fibres

179. During pregnancy, hormones like growth hormone, prolactin thyroxine, estrogen, progesterone *etc.*, increase several folds in the maternal blood. Which of the following are the reasons behind this increase in the hormone levels?

- a. To support foetal growth
- b. To support metabolic changes in the mother
- c. To maintain pregnancy
- d. To promote menstrual cycle on regular basis during pregnancy

Select the **correct** option.

- (1) a and b only
- (2) b and c only
- (3) a and d only
- (4) a, b and c only

180. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): In a 35 days menstrual cycle of human female, the day of ovulation will be 21st.

Reason (R): LH surge occurs on 9th day after the end of menstruation phase irrespective of the duration of the cycle. In the light of above statements, select the **correct** option.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (3) (A) is true but (R) is false
- (4) Both (A) and (R) are false



**Scan the QR Code to know
"Sleep or Leisure ?"**

